HADI HAMOUD

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SUMMARY

Applied AI engineer with 3+ years building, shipping and evaluating LLM-powered products. Own the evaluation loop end to end — benchmark and evaluation datasets, regression suites, reviewer workflows and A/B feedback — alongside RAG/retrieval over production corpora and agentic tool orchestration. Run fine-tuning data pipelines and self-hosted model serving on Kubernetes, working in Python, TypeScript and Go.

SKILLS

Applied AI & Agents: Vercel AI SDK, agent/tool orchestration, RAG and retrieval, MCP, prompt/context engineering, tool use and memory, knowledge graphs, vector search
LLM Evaluation: Araware, benchmark and evaluation datasets, regression tests, reviewer workflows, A/B feedback, model comparison and error analysis
LLMs & Serving: OpenAI, Anthropic, Gemini, Azure OpenAI, AWS Bedrock, Mistral, Groq/OpenRouter, HuggingFace, vLLM, TGI
Languages & Backend: Python, TypeScript/JavaScript, Node.js, Go, SQL, REST APIs, Next.js/React, n8n
Data & Infra: PostgreSQL/pgvector, Milvus, Chroma, ClickHouse, Redis, OpenSearch/BlackLab, Airflow, Docker, Kubernetes, CI/CD, OpenTelemetry, Grafana stack
Spoken: Arabic (native), English (fluent), French (intermediate, B1–B2)

EXPERIENCE

Founding Engineer Mar 2023 - Present
U4RASD - Arab Center for Research and Policy Studies Doha, Qatar

DALLA al Mojami — agentic Arabic linguistic assistant

- Built and shipped an Arabic conversational assistant over the Doha Historical Dictionary, covering meanings, roots, historical attestations, etymology, grammar, synonyms and corpus usage.
- Architected a controlled agentic pipeline with structured intent routing, parallel read-only retrieval over PostgreSQL and BlackLab, evidence-grounded generation, follow-up state, and a bounded specialist agent that validates and retries SQL queries.
- Built the full-stack product with Next.js, React, TypeScript, Vercel AI SDK, PostgreSQL and Drizzle; added provider adapters for hosted and self-hosted LLMs, Docker/CI delivery, Sentry/OpenTelemetry tracing, regression fixtures, reviewer datasets, smoke tests and A/B feedback.

Araware — LLM evaluation and benchmarking platform

- Built evaluation datasets for NER, relation extraction and Word Sense Disambiguation, and am building the orchestration platform spanning experiment management, model runners and a review dashboard.
- Designed and built an n8n workflow to automate benchmark execution: ingesting XLSX submissions, validating and routing direct and indirect task formats, running OpenAI chains with structured outputs, persisting results in PostgreSQL, exporting results and logging failures.

DALLA (Doha Arabic Large Language Models)

- Built end-to-end pretraining and finetuning data pipelines: provisioning, quality checking, custom stemming, deduplication, readability analysis and ranking.
- Implemented inference hosting using vLLM and HuggingFace TGI (cloud + on-premise).
- Developed the DALLA chatbot and its job-queuing/orchestration system; supported tokenizer experiments and data-generation pipelines for future releases.

iSOMAC — large-scale distributed data and NLP platform

- Designed and led a distributed collection platform with orchestration services, worker queues, scheduling, Kubernetes compute operators, versioned clients, and browser/API automation across six social platforms.
- Operated a 1000-pod Kubernetes deployment with automated scaling and monitoring.
- Built production NLP pipelines for NER, topic classification/modelling, dialect and language identification, and discourse analysis, exposed through a unified Information Extraction API.
- Built iSOMAC Agent, a natural-language interface over the platform's indexed data that answers queries and generates reports.

DevOps and Infrastructure

- Took on the Site Reliability Engineer role for the unit, supporting all projects with compute, hosting and infrastructure across on-premise and three main cloud providers.
- Supported migration of all core systems to a managed Kubernetes environment with CI/CD, and onboarded developers onto the new deployment and reliability workflows.
- Managed SQL, OLAP and object-storage environments; instrumented services using Grafana, Loki, Tempo, Mimir and Sentry.

Encyclopedia Arabica

- Set up multi-provider OCR tooling and supported large-scale data collection and a Wikidata-based knowledge-graph generation platform.

Leadership and Hiring

- Screened 2500+ resumes, interviewed 120 candidates and contributed to hiring 25 engineers; mentored 6 interns and coordinated project cycles.

Enterprise Solutions & Cloud Consultant Intern Jul 2021 - Oct 2021
KPMG Lower Gulf Limited Dubai, UAE

- Composed technical and commercial proposals and ERP deliverables for clients across multiple industries, including a CRM implementation for a major UAE government entity; participated in a full ERP selection and assessment cycle.

Fintech Software Engineering Intern Aug 2020 - Nov 2020
Areeba Beirut, Lebanon

- Designed and implemented a mobile push-notification migration for Bank Audi across Oracle DB, a UNIX dedicated server and the banking app; assisted with ATM setup and the implementation of VISA Direct OCT/AFT methods.

EDUCATION

American University of Beirut Aug 2018 - Aug 2022
Bachelor of Engineering in Computer and Communications

PUBLICATIONS

- Nacar et al. 2026. **DEAM: Diverse Embeddings for Arabic Multimodal Retrieval**. EMNLP 2026 (to appear).
- Chhaytle, **Hamoud**, et al. 2026. **Knowledge Graphs and Distributional Readability of Arabic Documents**. ArabicNLP 2026 (to appear).
- Ballout et al. 2026. **NeoAraBERT: A Modern Foundation Model for Arabic Embeddings**. Findings of ACL 2026.
- **Hamoud** et al. 2026. **NAKBA NLP 2026: Shared Task on Arabic Handwritten Manuscript Understanding**. LREC 2026.
- Bashiti et al. 2025. **ImageEval 2025: The First Arabic Image Captioning Shared Task**. ArabicNLP 2025.
- Mraikhat, **Hamoud**, Zaraket. 2024. **AREEj: Arabic Relation Extraction with Evidence**. ArabicNLP 2024.
- **Hamoud** et al. 2023. **DAVE: Differential Diagnostic Analysis Automation and Visualization from Clinical Notes**. EACL 2023 (System Demonstrations).
- Albared, **Hamoud**, Zaraket. 2023. **Arabic Topic Classification in the Generative and AutoML Era**. ArabicNLP 2023.
- Hamdan et al. 2024. **DRU at WojoodNER 2024: ICL LLM for Arabic NER**. ArabicNLP 2024.
- **Hamoud** et al. 2024. **DRU at WojoodNER 2024: A Multi-level Method Approach**. ArabicNLP 2024.